

Graduation Project Report

Comprehensive Layer 2 Security Implementation

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Chapter 1

Layer 2 Network Security Implementation and Mitigation

1.1 Introduction to Layer 2 Threats

1.1.1 Overview and Scope

This project report details the implementation and verification of critical Cisco Layer 2 security features to mitigate common threats such as MAC address flooding, VLAN attacks, and ARP/DHCP spoofing. The approach is based on common best practices and CCNA Security challenges. Layer 2 attacks historically account for 60-80% of common attacks in a LAN environment.

1.1.2 Key Vulnerabilities Addressed

- **CAM Table Overflow (MAC Address Flood):** Exploiting the Content-Addressable Memory table limit.
- **VLAN Hopping & Double Tagging:** Circumventing VLAN isolation.
- **DHCP Spoofing/Starvation:** Impersonating a DHCP server or exhausting the IP pool.
- **ARP Spoofing:** Manipulating ARP caches for Man-in-the-Middle (MITM) attacks.
- **STP Manipulation:** Forcing an attacker's device to become the Root Bridge.

1.2 Mitigation: Port Security against CAM Flooding

1.2.1 Explanation of Port Security

Port Security restricts the number of MAC addresses that can be learned on a specific switch port. This prevents an attacker from flooding the switch's CAM table by limiting the MAC entries per port.

Learning Methods

- **Static:** MAC address is manually hard-coded.
- **Dynamic:** MAC address is learned and stored in RAM (lost upon reboot).
- **Sticky:** MAC address is learned dynamically but saved to the running configuration, making it persistent.

Violation Modes

- **Shutdown (Default):** The port is disabled and placed in the `error-disable` state.
- **Restrict:** Violating traffic is dropped, a syslog message is generated, and the violation counter is incremented.
- **Protect:** Violating traffic is dropped without notifications or counter increments.

1.2.2 Configuration

```
1 Switch(config)# interface FastEthernet 0/1
2 Switch(config-if)# switchport mode access
3 Switch(config-if)# switchport port-security
4 Switch(config-if)# switchport port-security maximum 2
5 Switch(config-if)# switchport port-security mac-address sticky
6 Switch(config-if)# switchport port-security violation shutdown
7 ! Optional: Configure automatic recovery from error-disable
  state
8 Switch(config)# error-disable recovery cause psecurity
9 Switch(config)# error-disable recovery interval 30
```

Listing 1.1: Configuration for Port Security

1.2.3 Verification and Troubleshooting

```
1 # show port-security
2 # show port-security interface FastEthernet 0/1
3 # show mac address-table count
4 # show interfaces status err-disable
```

Listing 1.2: Verification Commands

1.3 VLAN Security: Hopping, Tagging, and PVLANS

1.3.1 VLAN Hopping Mitigation

The **Switch Spoofing** attack involves an attacker sending Dynamic Trunking Protocol (DTP) frames to force a port into a trunk link, granting access to multiple VLANs.

Configuration: Disabling DTP

```
1 Switch(config)# interface FastEthernet 0/2
2 Switch(config-if)# switchport mode access
3 Switch(config-if)# switchport nonegotiate ! Disables DTP
```

Listing 1.3: Mitigating VLAN Hopping

1.3.2 Double Tagging Mitigation

This is a one-way attack where two 802.1Q tags are used. The first switch removes the outer tag (if it's the Native VLAN), and the inner tag is used to reach the targeted VLAN.

Mitigation 1: Change Native VLAN

The primary mitigation is to change the **Native VLAN** from the default (VLAN 1) to an unused VLAN ID (e.g., VLAN 999).

```
1 Switch(config)# interface GigabitEthernet 0/1
2 Switch(config-if)# switchport trunk native vlan 999
```

Listing 1.4: Changing Native VLAN on Trunk Ports

Mitigation 2: Native VLAN Tagging (vlan dot1q)

An aggressive mitigation is to enforce 802.1Q tagging on the Native VLAN itself, preventing the switch from removing the outer tag, thus blocking the attack.

```
1 ! Note: This feature is not supported on all Cisco platforms.
2 Switch(config)# vlan dot1q tag native
```

Listing 1.5: Forcing Tagging on Native VLAN

1.3.3 Private VLANs (PVLANS)

PVLANS enhance isolation by restricting communication between hosts within the same Primary VLAN.

PVLAN Port Types

- **Primary VLAN:** The main VLAN that encapsulates the sub-VLANs.
- **Promiscuous Port:** Connects to the gateway/router and can communicate with all other ports (Isolated and Community).
- **Isolated Port:** Can only communicate with the Promiscuous port.
- **Community Port:** Can communicate with other Community ports and the Promiscuous port.

PVLAN Configuration Example

```

1  Switch(config)# vlan 100
2  Switch(config-vlan)# private-vlan primary
3  Switch(config-vlan)# private-vlan association 110,120 ! 110:
   Isolated, 120:Community
4  !
5  Switch(config)# vlan 110
6  Switch(config-vlan)# private-vlan isolated
7  !
8  Switch(config)# vlan 120
9  Switch(config-vlan)# private-vlan community
10 !
11 Switch(config)# interface FastEthernet 0/3 ! Example
   Promiscuous Port
12 Switch(config-if)# switchport mode private-vlan promiscuous
13 Switch(config-if)# switchport private-vlan mapping 100 110,120
14 !
15 Switch(config)# interface FastEthernet 0/4 ! Example Host Port
16 Switch(config-if)# switchport mode private-vlan host
17 Switch(config-if)# switchport private-vlan host-association
   100 110

```

Listing 1.6: Private VLAN Configuration Steps

1.3.4 Verification (VLAN)

```

1  # show vlan brief
2  # show vlan private-vlan
3  # show interface FastEthernet 0/4 switchport

```

Listing 1.7: Verification Commands for VLANs and PVLANs

1.4 Mitigation: DHCP Snooping and DAI

1.4.1 DHCP Snooping for Spoofing and Starvation

DHCP Spoofing occurs when a rogue DHCP server provides false IP information. **DHCP Starvation** is a DoS attack that consumes the IP pool.

DHCP Snooping is a Layer 2 security feature that operates by classifying ports as **Trusted** or **Untrusted**. The port connected to the legitimate DHCP server is set to **Trusted**, while end-user ports are **Untrusted**.

Its primary role is to inspect DHCP messages passing through the switch, dropping any DHCP response (Offer/ACK) coming from an **Untrusted** port (i.e., a rogue DHCP server). It also creates a **Binding Table** linking the client's MAC, IP, and VLAN, which is subsequently utilized by ****Dynamic ARP Inspection (DAI)**** for enhanced security.

Configuration (DHCP Snooping)

```
1      ! Enable DHCP snooping globally and specify VLANs
2      Switch(config)# ip dhcp snooping
3      Switch(config)# ip dhcp snooping vlan 10,20
4      !
5      ! Enable Option 82 (DHCP Relay Information) to verify source
6      Switch(config)# ip dhcp snooping information option
7      !
8      ! Set the port connected to the DHCP server as trusted
9      Switch(config)# interface GigabitEthernet 0/2
10     Switch(config-if)# ip dhcp snooping trust
11     !
12     ! Rate limit untrusted client ports to prevent starvation
13     Switch(config)# interface FastEthernet 0/5
14     Switch(config-if)# ip dhcp snooping limit rate 5
```

Listing 1.8: DHCP Snooping Configuration

1.4.2 Dynamic ARP Inspection (DAI) for ARP Spoofing

ARP Spoofing uses forged ARP replies to associate an attacker's MAC address with a legitimate IP address.

DAI validates ARP packets against the DHCP Snooping binding table, dropping any invalid ARP traffic from untrusted ports.

Configuration (DAI)

```
1      ! Enable DAI on the relevant VLANs (DAI requires DHCP snooping
2      ! to be active)
3      Switch(config)# ip arp inspection vlan 10,20
4      !
5      ! Set the uplink/router port as trusted
6      Switch(config)# interface GigabitEthernet 0/2
7      Switch(config-if)# ip arp inspection trust
```

Listing 1.9: Dynamic ARP Inspection Configuration

1.4.3 IP Source Guard (IPSG)

IP Source Guard (IPSG) is a security feature that leverages the DHCP Snooping binding table to restrict non-ARP traffic. IPSG prevents a host from using an unauthorized source IP address or MAC address, effectively blocking spoofing attempts at the IP level. It acts as a final layer of defense by verifying the source IP and MAC against the trusted binding table.

Configuration (IPSG)

```
1      ! Enable IP Source Guard on untrusted ports using the 'ip
      verify source' command
2      Switch(config)# interface FastEthernet 0/5
3      Switch(config-if)# ip verify source
```

Listing 1.10: IP Source Guard Configuration

1.4.4 Verification and Troubleshooting

```
1      ! Verification for DHCP Snooping Binding Table (used by DAI
      and IPSG)
2      # show ip dhcp snooping binding
3      !
4      ! Verification for DAI and IPSG statistics
5      # show ip arp inspection statistics
6      # show ip verify source
```

Listing 1.11: Verification Commands for DHCP/ARP/IPSG

1.5 STP Security and Storm Control

1.5.1 STP Attacks and Root Guard

STP attacks involve injecting superior BPDUs to make the attacker's device the Root Bridge.

Root Guard prevents a port from becoming a Root Port. If a superior BPDU is received on a Root Guard enabled port, the port is placed in a **root-inconsistent** state, blocking traffic.

Configuration (Root Guard)

```
1      Switch(config)# interface GigabitEthernet 0/3
2      Switch(config-if)# spanning-tree guard root
```

Listing 1.12: Root Guard Configuration

1.5.2 BPDU Guard

BPDU Guard is enabled on access ports connected to end-hosts which should never receive BPDUs. If a BPDU is received, the port is instantly shut down to the **error-disable** state.

Configuration (BPDU Guard)

```
1      ! Globally enable PortFast and BPDU Guard on all PortFast-
      enabled interfaces
2      Switch(config)# spanning-tree portfast default
```



```
3 Switch(config)# spanning-tree bpduguard enable
```

Listing 1.13: BPDU Guard Configuration (Global)

1.5.3 Loop Guard (Guard Loop)

Loop Guard prevents Spanning Tree Protocol (STP) ports that are designated to be in a blocking state from transitioning to a forwarding state. This occurs if the port stops receiving BPDUs, typically due to software or hardware failure, which could otherwise create an accidental network loop.

Configuration (Loop Guard)

```
1 Switch(config)# interface GigabitEthernet 0/4
2 Switch(config-if)# spanning-tree guard loop
```

Listing 1.14: Loop Guard Configuration (Per Interface)

1.5.4 UniDirectional Link Detection (UDLD)

UDLD monitors the physical configuration of fiber and copper links to detect unidirectional links, which can cause subtle network errors and STP failures. UDLD has two modes: **Normal** (disables the port when a unidirectional link is detected) and **Aggressive** (shuts down the port immediately if UDLD packets are not received within a short window).

Configuration (UDLD)

```
1 ! Enable UDLD globally
2 Switch(config)# udld enable
3
4 ! Set aggressive mode on a specific interface
5 Switch(config)# interface FastEthernet 0/5
6 Switch(config-if)# udld aggressive
```

Listing 1.15: UDLD Configuration (Global and Interface)

1.5.5 Storm Control

Storm Control limits the excessive flow of broadcast, multicast, or unknown unicast traffic to prevent network slowdowns or failures.

Configuration (Storm Control)

```
1 Switch(config)# interface FastEthernet 0/6
2 Switch(config-if)# storm-control broadcast level 0.10 ! Limit
   broadcast to 10\%
3 Switch(config-if)# storm-control action shutdown
```

Listing 1.16: Storm Control Configuration

1.5.6 Verification (STP and Storm Control)

```
1 # show spanning-tree interface FastEthernet 0/1 detail
2 # show uddl interface FastEthernet 0/5
3 # show storm-control FastEthernet 0/6
```

Listing 1.17: Verification Commands